



Politecnico di Milano

Dipartimento di Elettronica, Informazione e Bioingegneria

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Parallel Computing– 1st mid-term test– Tuesday, November 12th 2024

Polimi ID _____

Surname _____ **Name** _____

- This is a closed-book examination. You cannot use computers, phones, or laptops during the exam.
- Paper will be provided, but you should bring and use writing instruments that yield marks dark enough to be read easily. Erasable pens can be used.
- Total available time: 1h:30m.

Exercise 1 (4 points) _____

Exercise 2 (4 points) _____

Exercise 3 (4 points) _____

Exercise 3 (4 points) _____

Exercise n. 1

Answer the following questions and briefly explain (without an explanation, the answer will be considered invalid)

- A. Please discuss the lemma deriving the slow-down factor when we move from an unbounded memory PRAM model to a bounded one. (1)

- B. Complete the following sentence and explain what it means: "In a PRAM model, a **read conflict** occurs when..." (1)

- C. Give a definition of the speedup when a PRAM model is considered. (1)

- D. Please outline the key aspects of Amdahl's Law. (1)

Exercise n. 2

Answer the following questions and briefly explain (without an explanation, the answer will be considered invalid).

- A. Show an example where, in a data-parallel model, the SIMD execution is not reaching the maximum efficiency. (2)

- B. List what a programmer can do to reduce the slow-down due to the memory stalls. (1)

- C. "In a superscalar processor, the parallelism is automatically discovered." True/False? Why?(1)

Exercise n. 3

Answer the following questions and briefly explain (without an explanation, the answer will be considered invalid).

- A. Please describe the different hardware architectures supporting multi-threading. (1)

- B. Please describe how a programmer can reduce the penalties due to the memory bandwidth bound. (1)

- C. Describe the types of address spaces CUDA kernels work with. How many? How much is shared? How fast are they? (2)

Exercise n. 4

Answer the following questions and briefly explain (without an explanation, the answer will be considered invalid).

- A. What is a mutex in Pthreads, and how does it work? (1)

- B. In OpenMP, tasks are better than sections for irregular algorithms. Is this statement true or false? Why? (1) **This question can be skipped if you got 1 or 3 points in the first challenge.**

- C. The following code snippet is supposed to always run Task 3 after Tasks 1 and 2. Why does it sometimes not happen? (2) **This question can be skipped if you got 2 or 3 points in the first challenge.**

```
1 #pragma omp parallel {  
2   #pragma omp single {  
3     #pragma omp task { Task1(); }  
4     #pragma omp task { Task2(); }  
5     Task3();  
6   }  
7 }
```
